

Motivation

Scientific gap

VFSMOD: linear sorption only

Objective

Add Freundlich isotherm, compare

Scope

Koc 1-10⁶ L/kg, t½ 1-300 d

Code

COMMON/WQ1

+VKF(10), VKN(10) · 6 files

CDE leaching (cde.f)

$R = f(N, Kf, C)$ via N-R iteration

Partitioning (wqpest.f)

$s = Kf \cdot C^N$ · N-R residue split

Benchmark (VFSMOD v4.6.2)

sampleP dP=64.09% · samplePF dP=64.46% · IKD=0,1,2 backward compatible

Scenarios

Multi-event framework

3 events · mresn carry-over

2 isotherms

Linear (Kd) · Freundlich (Kf=Kd, N=0.85)

2 application modes

All events · single application

Koc × t½ sweep (Fig. 3)

Koc 1-10⁶ L/kg · t½ 1-300 d · 4 scenario combinations → mresn, mF, mfml sweet-spot curves

GSA

Saltelli sampling

$L=512$ · $k=6$ · $n=4096/\text{scenario}$

6 parameters

Koc · N · t½ · VL · DM · Ks

Parallel execution

49,152 VFSMOD calls · 7 cores

Jansen S1 & ST indices (Fig. 5)

t½ dominates mresn (ST=0.82-0.93) · DM second (ST=0.66-0.80)
Ks controls dP, dQ (ST≈1) · Koc near-zero (non-monotonic response)

Results

Sweet-spot shift (Fig. 4)

Koc ~30→300 · peak up to 22×

Residue amplification

3-17× mresn · Fre/Lin ratio

Regulatory context

FOCUS VFS · SWAN coupling

Main finding

t½ and DM drive residue fate · Freundlich amplifies both drivers
Koc 100-1000 L/kg most sensitive to isotherm choice

Outputs

Figs. 1-5 · VFSMOD v4.6.2 · SETAC 2026 abstract · GSA dataset (49,152 runs)